Wasserstein convergence rate for stationary finite-state vector Markov additive functionals, allowing periodicity

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Editorial note. The fixed source is arXiv2401.15719v5 dated2026-08-29. Source123 explicitly states that this version corrects errors in the Math of OR published version; do not substitute a prior version or reuse its proof silently. The officialv5 source archive names its main filearxiv-v6.tex, which is an internal filename rather than the arXiv version number.

Editorial note. Only the finite-state stationary theorem946-962 is counted. Periodicity is allowed; the separate arbitrary-initial-distribution corollary and its aperiodicity assumption are not included in this claim.

Editorial note. The already public abstract martingale theorem199-281 and its complete proof are included as supporting solution obligations. No new count is assigned to that theorem or to the Hessian-variation lemma separately.

Editorial note. The source uses the same letterh when passing through the Lipschitz rescaling bijection305-311 and taking the supremum at738. The final theorem is a supremum statement; the review must preserve that rescaling convention rather than identify the unscaled and normalized individual test functions.

Editorial scope. One stationary finite-state vector MarkovWassersteinCLT and its optimized log-squared rate, with all local core retained. The general-state extension and TD-learning theorem have not been fully prepared by this card.

Allowed standard prerequisites (editorial).

- Finite irreducible Markov-chain Poisson equation. For a finite irreducible transition matrixP and a centered functiong withpi(g)=0, (I-P)V=g has a bounded solution, also for a periodic chain; apply entrywise to centered matrix functions.. Produce reward and covariance potentials. No quantitative Markov CLT is supplied.

- Standard Gaussian Stein solution regularity. For a Lipschitz test function, the normal Stein solution has bounded Hessian and beta-Holder Hessian with constants of order(1-beta)^(-1), as explicitly stated in source.. Bound the two martingale Taylor errors. The local martingale bound, its signed covariance term and its control are proof obligations.

- Gaussian integration by parts, Mehler formula and smoothing derivative estimates. ForP_Ah(x)=E[h(x+A^(1/2)Z)] andh1-Lipschitz, ||nabla^j P_Ah||<=C_d||A^(-1/2)||^(j-1), j=2,3,4; use the Gaussian covariance interpolation identity.. Derive the source Hessian-weight representation and separately control shift/internal-covariance/external-covariance changes.
- Martingale conditioning, finite-dimensional matrix inequalities and summation by parts. No independence between the random Stein Hessian and the Markov conditional covariance is assumed.

0.1 Preliminaries

In this subsection, we will describe the notation we use to establish the rate of convergence in the central limit theorem for vector-valued martingale differences. Let $\{m_k\}_{k\geq 1}$ be a d-dimensional martingale difference sequence with respect to a filtration $\{\mathcal{F}_k\}_{k\geq 0}$ i.e., m_k is $\mathcal{F}_k$-adapted, $E(||m_k||) < \infty$, where $||\cdot||$ is the standard Euclidean norm in $\Re^d$ and

$$E(m_k|\mathcal{F}_{k-1}) = 0 \qquad \forall k \geq 1.$$

Assumption 1. *Fix $\beta \in (0,1)$. We assume that the martingale difference sequence satisfies the following properties:*

1. *$E\|m_k\|^{2+\beta} < \infty$ for every $k \geq 1$.*
2. *$\Sigma_k = E(m_k m_k^T \mid \mathcal{F}_{k-1})$ exists for every $k \geq 1$.*

$\diamond$

Item (1) in the assumption is motivated by the fact that we want to extend the techniques in (Gallouët et al. 2018) to martingales .

Define $S_n = \sum_{k=1}^n m_k$. We are interested in whether the following normalized sum

$$W_n := \frac{S_n}{\sqrt{n}} \tag{1}$$

converges to $N(0, \Sigma_\infty)$, and if so, we are interested in the rate of convergence. We will use Stein's method to establish such a convergence and to estimate the rate of convergence (Stein 1972, Ross 2011, Chatterjee 2014).

The Wasserstein distance between two d-dimensional random vectors X and Y is defined as

$$d_{\mathcal{W}}(X,Y) = \sup_{h\in Lip_1} E[h(X) - h(Y)],$$

where Lip_L, the class of Lipschitz functions from $\Re^d$ to $\Re$ with Lipschitz constant L, is defined as

$$Lip_L = \{h : |h(x) - h(y)| \leq L||x-y||\}.$$

For later reference, we use the notation $||A||_{op}$ to denote the operator norm of a matrix A, i.e.,

$$||A||_{op} = \sup_{x:||x||=1} ||Ax||.$$

The following results from (Gallouët et al. 2018, Propositions 2.2 and 2.3) and (Fang et al. 2019, Theorem 3.1) will be useful to us and so we present them together as a lemma for future reference.

Lemma 1. *Let $g \in Lip_L$ and f be the solution to Stein's equation:*

$$g(u) - E(g(Z)) = \Delta f(u) - u^T \nabla f(u), \tag{2}$$

where Δ and ∇ are the Laplacian and gradient operators, respectively, and $Z \sim N(0, I)$. Then, f has the following regularity properties for any $\beta \in (0, 1)$:

1. $||\nabla^2 f(x) - \nabla^2 f(y)||_{op} \leq \tilde{C}_1(d, \beta)||x - y||^\beta L$, *where ∇^2 is the Hessian operator.*
2. $|\Delta f(x) - \Delta f(y)| \leq \tilde{C}_2(d, \beta)||x - y||^\beta L$, *where*

$$\tilde{C}_1 = C_1(d) + \frac{2}{1-\beta}, \quad \tilde{C}_2 = C_2(d) + \frac{2d}{1-\beta}, \quad C_1 = 2^{3/2}\frac{(1+2d)\Gamma((1+d)/2)}{d\Gamma(d/2)}, \quad C_2 = 2\sqrt{\frac{2d}{\pi}}.$$

3. $$\left|\langle \nabla^2 f(x), u_1 u_2^T \rangle_{HS}\right| \leq CL||u_1|| \cdot ||u_2|| \, \forall x, u_1, u_2 \in \Re^d,$$

   where C is a constant (independent of d and β) and $\langle \cdot, \cdot \rangle_{HS}$ is the Hilbert-Schmidt inner product (also called the Frobenius inner product), i.e., the sum of the element-by-element product of two matrices.

$\diamond$

1 Markov Chain Central Limit Theorem

Consider a time-homogeneous Markov chain $\{X_k\}_{k \geq 0}$ on a state space $\mathcal{S}$ with initial distribution η. Let $r(X_k) \in \Re^d$ denote a vector reward and let $\bar{r} = \pi r$ whenever the chain has invariant distribution π. We are interested in rates of convergence to normality of the centered additive functional

$$\frac{1}{\sqrt{n}} \sum_{k=1}^{n} \big(r(X_{k-1}) - \bar{r}\big).$$

As is common in the literature, we study this scaled sum by relating the reward process to a martingale via Poisson's equation (Douc et al. 2018, Makowski and Shwartz 2002, Glynn and Meyn 1996, Glynn and Infanger 2023).

1.1 Finite State Space Markov Chains

We now show that the martingale theorem can be used to obtain a non-asymptotic CLT for finite-state Markov chains. In Theorem 2, one must control the signed covariance-interaction term. This can be done by applying a second Poisson equation, now to the conditional covariance function.

Throughout this subsection, assume that the Markov chain is stationary, finite-state, and irreducible. Periodicity is allowed. Let

$$g(x) := r(x) - \bar{r}, \qquad \bar{r} := \pi r.$$

Let V solve the Poisson equation

$$V - PV = g.$$

Then

$$g(X_{k-1}) = V(X_{k-1}) - V(X_k) + m_k,$$

where

$$m_k := V(X_k) - E[V(X_k) \mid X_{k-1}]$$

is a martingale difference sequence. Define

$$\Sigma_k := E[m_k m_k^\top \mid \mathcal{F}_{k-1}] = v(X_{k-1}),$$

where

$$v(x) := E[m_1 m_1^\top \mid X_0 = x].$$

Since the chain is stationary,

$$E(\Sigma_k) = \pi(v) =: \Sigma_\infty.$$

Assume throughout this subsection that Σ_∞ is positive definite.

Theorem 1. *Let*

$$U_n = \frac{1}{\sqrt{n}} \sum_{k=1}^{n} (r(X_{k-1}) - \bar{r}).$$

Assume that the Markov chain is stationary, finite-state, and irreducible, and that $\Sigma_\infty \succ 0$. Then, for every $\beta \in (0,1)$,

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \le \frac{C}{(1-\beta)^2} n^{-\beta/2},$$

where C does not depend on n or β. Consequently,

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) = O\left(\frac{(\log n)^2}{\sqrt{n}}\right).$$

Theorem 2. *Let*

$$V_k := \sum_{j=k}^{n} E(\Sigma_j), \qquad k = 1, \dots, n,$$

and assume that V_k is positive definite for each $k = 1, \dots, n$. Define

$$D_k := V_k^{-1/2} E(\Sigma_k) V_k^{-1/2}, \qquad k = 1, \dots, n.$$

Let $Z \sim N(0, I)$, and let W_n be defined as in (1). Then, under Assumption 1 for the same fixed $\beta \in (0,1)$,

$$d_{\mathcal{W}}(W_n, n^{-1/2} V_1^{1/2} Z) \le \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \Bigg[\tilde{C}_1(d,\beta) \|V_k^{1/2}\|_{op} E\left(\|V_k^{-1/2} m_k\|^{2+\beta}\right) + \tilde{C}_1(d,\beta) \operatorname{Tr}(D_k) \|V_k^{1/2}\|_{op} E\left(\|V_k^{-1/2} m_k\|^{\beta}\right) \Bigg] + \Delta_n,$$

where

$$\Delta_n := \sup_{h \in Lip_1} \frac{1}{\sqrt{n}} \left| \sum_{k=1}^{n} E\left[\operatorname{Tr}\left(A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2} \left(E(\Sigma_k) - \Sigma_k\right) V_k^{-1/2} \right) \right] \right|.$$

Here

$$A_{k,h}(\mathcal{F}_{k-1}) := E\left[\nabla^2 f_{k,h}(\widetilde{Z}_k) \mid \mathcal{F}_{k-1}\right],$$

where $f_{k,h}$ is the $\mathcal{F}_{k-1}$-measurable random solution of the Stein equation defined in the proof below with

$$a = S_{k-1}, \qquad Q_k = V_k^{1/2}.$$

and

$$\widetilde{Z}_k := V_k^{-1/2} T_{k+1} \sim N\left(0, V_k^{-1/2} V_{k+1} V_k^{-1/2}\right),$$

with the convention $V_{n+1} = 0$. Moreover,

$$\|A_{k,h}(\mathcal{F}_{k-1})\|_{op} \le C \|V_k^{1/2}\|_{op}, \quad a.s.,$$

where C is the constant in statement (3) of Lemma 1. Since $0 \preceq D_k \preceq I$, the same bound also implies the coarser version obtained by replacing $\mathrm{Tr}(D_k)$ by d. The sharper form above is used in the Markov-chain applications.

Proof. Fix $h \in Lip_1$. By the standard rescaling of Lipschitz test functions,

$$d_{\mathcal{W}}(W_n, n^{-1/2} V_1^{1/2} Z) = \sup_{h \in Lip_1} \frac{1}{\sqrt{n}} E\left[h(S_n) - h(T_1)\right],$$

where

$$T_k := \sum_{j=k}^{n} Y_j,$$

and $Y_1, \ldots, Y_n$ are independent Gaussian random vectors, independent of the martingale difference sequence, with

$$Y_j \sim N(0, E(\Sigma_j)).$$

Thus

$$T_k \sim N(0, V_k).$$

The rescaling follows because if $g \in Lip_1$, then $h(x) := \sqrt{n}\, g(x/\sqrt{n})$ also belongs to Lip_1, and

$$E\left[g(W_n) - g(n^{-1/2} V_1^{1/2} Z)\right] = \frac{1}{\sqrt{n}} E\left[h(S_n) - h(T_1)\right].$$

Using the Lindeberg decomposition,

$$E\left[h(S_n) - h(T_1)\right] = \sum_{k=1}^{n} E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k)\right],$$

where $T_{n+1} = S_0 = 0$.

We now bound a single summand. Fix k. Conditional on $\mathcal{F}_{k-1}$, set

$$a = S_{k-1}, \qquad Q_k = V_k^{1/2}.$$

Define

$$\widetilde{h}_k(u) := h(a + Q_k u).$$

Since $h \in Lip_1$, we have

$$\widetilde{h}_k \in Lip_{L_k}, \qquad L_k = \|Q_k\|_{op} = \|V_k^{1/2}\|_{op}.$$

Let $f_k = f_{k,h}$ be the solution of the Stein equation

$$\widetilde{h}_k(u) - E[\widetilde{h}_k(Z)] = \Delta f_k(u) - u^\top \nabla f_k(u).$$

Since $a = S_{k-1}$ is $\mathcal{F}_{k-1}$-measurable, the function f_k is also $\mathcal{F}_{k-1}$-measurable as a random function.

By the Stein equation, with

$$\widetilde{Z}_k := V_k^{-1/2} T_{k+1},$$

we have

$$\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&= E\left[\Delta f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) - \left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)^\top \nabla f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) \mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

Moreover,

$$\widetilde{Z}_k \sim N(0, \Lambda_k), \qquad \Lambda_k := V_k^{-1/2} V_{k+1} V_k^{-1/2}.$$

Since

$$V_k = E(\Sigma_k) + V_{k+1},$$

we have

$$I - \Lambda_k = V_k^{-1/2} E(\Sigma_k) V_k^{-1/2}.$$

We rewrite the preceding conditional expectation as

$$\begin{aligned}
&E\left[\mathrm{Tr}\left(\Lambda_k \nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) - \widetilde{Z}_k^\top \nabla f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) \mid \mathcal{F}_{k-1}\right] \\
&+ E\left[\mathrm{Tr}\left((I - \Lambda_k) \nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) \mid \mathcal{F}_{k-1}\right] \\
&- E\left[(V_k^{-1/2} m_k)^\top \nabla f_k(\widetilde{Z}_k) \mid \mathcal{F}_{k-1}\right] \\
&- E\left[(V_k^{-1/2} m_k)^\top \left(\nabla f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) - \nabla f_k(\widetilde{Z}_k)\right) \mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

The first term is zero by Gaussian integration by parts: conditional on $\mathcal{F}_k$, the quantity $V_k^{-1/2} m_k$ is fixed and $\widetilde{Z}_k \sim N(0, \Lambda_k)$ is independent of $\mathcal{F}_k$. The third term is zero because f_k is $\mathcal{F}_{k-1}$-measurable as a random function, $\widetilde{Z}_k$ is independent of $\mathcal{F}_k$, and $E[m_k \mid \mathcal{F}_{k-1}] = 0$.

Using the fundamental theorem of calculus, we obtain

$$\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&= E\left[\mathrm{Tr}\left(V_k^{-1/2} E(\Sigma_k) V_k^{-1/2} \nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) \mid \mathcal{F}_{k-1}\right] \\
&\quad - E\left[(V_k^{-1/2} m_k)^\top \left(\nabla^2 f_k(\Theta V_k^{-1/2} m_k + \widetilde{Z}_k) - \nabla^2 f_k(\widetilde{Z}_k)\right)(V_k^{-1/2} m_k) \mid \mathcal{F}_{k-1}\right] \\
&\quad - E\left[(V_k^{-1/2} m_k)^\top \nabla^2 f_k(\widetilde{Z}_k)(V_k^{-1/2} m_k) \mid \mathcal{F}_{k-1}\right],
\end{aligned}$$

where $\Theta \sim \text{Uniform}[0,1]$ is independent of all other random variables.

Define

$$A_{k,h}(\mathcal{F}_{k-1}) := E\left[\nabla^2 f_k(\widetilde{Z}_k) \mid \mathcal{F}_{k-1}\right].$$

Using

$$\Sigma_k = E[m_k m_k^\top \mid \mathcal{F}_{k-1}],$$

the last term becomes

$$-\operatorname{Tr}\left(V_k^{-1/2}\Sigma_k V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1})\right).$$

By statement (1) of Lemma 1, the second term above is bounded in absolute value by

$$\tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2} m_k\|^{2+\beta} \mid \mathcal{F}_{k-1}\right].$$

Therefore,

$$\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&\le E\left[\operatorname{Tr}\left(V_k^{-1/2} E(\Sigma_k) V_k^{-1/2} \nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) \mid \mathcal{F}_{k-1}\right] \\
&\quad - \operatorname{Tr}\left(V_k^{-1/2}\Sigma_k V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1})\right) \\
&\quad + \tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2} m_k\|^{2+\beta} \mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

Next, add and subtract

$$\operatorname{Tr}\left(V_k^{-1/2} E(\Sigma_k) V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1})\right).$$

This gives

$$\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&\le \operatorname{Tr}\left(A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2}\left(E(\Sigma_k) - \Sigma_k\right) V_k^{-1/2}\right) \\
&\quad + E\left[\operatorname{Tr}\left(D_k\left[\nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) - \nabla^2 f_k(\widetilde{Z}_k)\right]\right) \mid \mathcal{F}_{k-1}\right] \\
&\quad + \tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2} m_k\|^{2+\beta} \mid \mathcal{F}_{k-1}\right],
\end{aligned}$$

where

$$D_k := V_k^{-1/2} E(\Sigma_k) V_k^{-1/2}.$$

It remains to bound the weighted trace term. Since

$$V_k = E(\Sigma_k) + V_{k+1},$$

we have

$$D_k = I - V_k^{-1/2} V_{k+1} V_k^{-1/2}.$$

Hence

$$0 \preceq D_k \preceq I.$$

Therefore, for any $x, u \in \mathbb{R}^d$,

$$\begin{aligned}
&\left|\mathrm{Tr}\left(D_k\left[\nabla^2 f_k(x+u) - \nabla^2 f_k(x)\right]\right)\right| \\
&\leq \mathrm{Tr}(D_k)\left\|\nabla^2 f_k(x+u) - \nabla^2 f_k(x)\right\|_{op} \\
&\leq \mathrm{Tr}(D_k)\,\tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op}\|u\|^\beta.
\end{aligned}$$

Applying this bound with

$$x = \widetilde{Z}_k, \qquad u = V_k^{-1/2} m_k,$$

and taking conditional expectation gives

$$\begin{aligned}
&\left|E\left[\mathrm{Tr}\left(D_k\left[\nabla^2 f_k\left(V_k^{-1/2}m_k + \widetilde{Z}_k\right) - \nabla^2 f_k(\widetilde{Z}_k)\right]\right) \mid \mathcal{F}_{k-1}\right]\right| \\
&\leq \tilde{C}_1(d,\beta)\,\mathrm{Tr}(D_k)\|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2}m_k\|^\beta \mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

Thus,

$$\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&\leq \mathrm{Tr}\left(A_{k,h}(\mathcal{F}_{k-1})V_k^{-1/2}\left(E(\Sigma_k) - \Sigma_k\right)V_k^{-1/2}\right) \\
&\quad + \tilde{C}_1(d,\beta)\,\mathrm{Tr}(D_k)\|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2}m_k\|^\beta \mid \mathcal{F}_{k-1}\right] \\
&\quad + \tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2}m_k\|^{2+\beta} \mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

Taking expectations, summing over k, substituting into the Lindeberg decomposition, and using the rescaling identity for the Wasserstein distance, we obtain, for the fixed $h \in Lip_1$,

$$\begin{aligned}
&E\left[h(W_n) - h(n^{-1/2}V_1^{1/2}Z)\right] \\
&\leq \frac{1}{\sqrt{n}}\sum_{k=1}^{n}\Bigg[\tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op}E\left(\|V_k^{-1/2}m_k\|^{2+\beta}\right) \\
&\qquad\qquad + \tilde{C}_1(d,\beta)\,\mathrm{Tr}(D_k)\|V_k^{1/2}\|_{op}E\left(\|V_k^{-1/2}m_k\|^\beta\right)\Bigg] \\
&\quad + \frac{1}{\sqrt{n}}\left|\sum_{k=1}^{n} E\left[\mathrm{Tr}\left(A_{k,h}(\mathcal{F}_{k-1})V_k^{-1/2}\left(E(\Sigma_k) - \Sigma_k\right)V_k^{-1/2}\right)\right]\right|.
\end{aligned}$$

Taking the supremum over $h \in Lip_1$ proves the claimed bound.

Finally, statement (3) of Lemma 1 gives

$$\|A_{k,h}(\mathcal{F}_{k-1})\|_{op} \leq C\|V_k^{1/2}\|_{op}.$$

This completes the proof. □

We first state the technical estimate needed to control the covariance interaction term.

Lemma 2. *Let*

$$V_k := \sum_{j=k}^{n} E(\Sigma_j) = (n-k+1)\Sigma_\infty, \qquad k = 1, \dots, n,$$

and

$$H_k := V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2},$$

where

$$A_{k,h}(\mathcal{F}_{k-1}) := E[\nabla^2 f_{k,h}(\widetilde{Z}_k) \mid \mathcal{F}_{k-1}]$$

is the matrix appearing in Theorem 2. Then, for every $\beta \in (0,1)$,

$$\sup_{h \in Lip_1} \sum_{k=2}^{n} E\|H_k - H_{k-1}\|_{op} \le \frac{C}{(1-\beta)^2} n^{(1-\beta)/2},$$

where C depends on the chain, r, d, and Σ_∞, but not on n or β.

Proof. Fix $h \in Lip_1$. Let $c_t := \sqrt{1-e^{-2t}}$, and let

$$P_A h(x) := E[h(x + A^{1/2} Z)], \qquad Z \sim N(0, I),$$

denote Gaussian convolution with covariance A. The Mehler representation of the standard normal Stein solution, followed by differentiation under the integral, gives

$$H_k = -\int_0^\infty e^{-2t} E_{\xi \sim N(0, V_{k+1})}[\nabla^2 P_{c_t^2 V_k} h(S_{k-1} + e^{-t}\xi)]\, dt. \tag{3}$$

The differentiation is justified by the standard Gaussian-smoothing bounds

$$\|\nabla^j P_A h\|_{op} \le C \|A^{-1/2}\|_{op}^{j-1}, \qquad j = 2, 3, 4.$$

Here $\|\cdot\|_{op}$ denotes the usual matrix operator norm for matrices and Hessians, and the multilinear operator norm for high-order derivatives.

The second- and third-derivative bounds imply, respectively,

$$\|\nabla^2 P_A h(x) - \nabla^2 P_A h(y)\|_{op} \le 2C \|A^{-1/2}\|_{op},$$
$$\|\nabla^2 P_A h(x) - \nabla^2 P_A h(y)\|_{op} \le C \|A^{-1/2}\|_{op}^2 \|x - y\|.$$

Since $\min\{1, u\} \le u^\beta$ for $u \ge 0$, interpolation gives

$$\|\nabla^2 P_A h(x) - \nabla^2 P_A h(y)\|_{op} \le C \|A^{-1/2}\|_{op}^{1+\beta} \|x - y\|^\beta, \tag{4}$$

with C uniform over $\beta \in (0,1)$.

Because the state space is finite, m_k is uniformly bounded. Moreover,

$$c(n-k+1)I \preceq V_k \preceq C(n-k+1)I.$$

Using (3), compare H_k and H_{k-1} by changing separately the spatial shift, the covariance inside the Gaussian convolution, and the covariance of the outer Gaussian average. Denote the corresponding contributions by $\Delta_{\mathrm{shift},k}$, $\Delta_{\mathrm{int},k}$, and $\Delta_{\mathrm{ext},k}$.

Since $S_{k-1} - S_{k-2} = m_{k-1}$, (4) gives

$$E\|\Delta_{\text{shift},k}\|_{op} \le CE\|m_{k-1}\|^{\beta} \int_0^{\infty} e^{-2t} \|(c_t^2 V_k)^{-1/2}\|_{op}^{1+\beta} dt$$
$$\le \frac{C}{1-\beta}(n-k+1)^{-(1+\beta)/2},$$

where

$$\int_0^{\infty} e^{-2t} c_t^{-(1+\beta)} dt = \frac{1}{1-\beta}.$$

For the internal covariance change, stationarity gives $V_{k-1} - V_k = \Sigma_\infty$. Along $A_s = c_t^2(V_k + s\Sigma_\infty)$, the Gaussian-semigroup identity and the fourth-derivative bound yield

$$E\|\Delta_{\text{int},k}\|_{op} \le C \int_0^{\infty} e^{-2t} c_t^2 \|(c_t^2 V_k)^{-1/2}\|_{op}^3 \|\Sigma_\infty\|_{op}\, dt$$
$$\le C(n-k+1)^{-3/2} \int_0^{\infty} e^{-2t} c_t^{-1} dt$$
$$\le C(n-k+1)^{-3/2}.$$

For the external covariance change, couple

$$\xi_{k-1} = \xi_k + \eta_k, \qquad \xi_k \sim N(0, V_{k+1}), \quad \eta_k \sim N(0, \Sigma_\infty),$$

with independent summands. Applying (4) to the additional shift $e^{-t}\eta_k$ gives

$$E\|\Delta_{\text{ext},k}\|_{op} \le \frac{C}{1-\beta}(n-k+1)^{-(1+\beta)/2}.$$

Consequently,

$$E\|H_k - H_{k-1}\|_{op} \le \frac{C}{1-\beta}(n-k+1)^{-(1+\beta)/2} + C(n-k+1)^{-3/2}.$$

Finally,

$$\sum_{q=1}^{n-1} q^{-(1+\beta)/2} \le \frac{C}{1-\beta} n^{(1-\beta)/2},$$

whereas $\sum_{q\ge1} q^{-3/2} < \infty$. Summing over k and taking the supremum over $h \in Lip_1$ proves the result. □

Proof. The reward Poisson equation gives

$$U_n = W_n + \frac{V(X_0) - V(X_n)}{\sqrt{n}}, \qquad W_n := \frac{1}{\sqrt{n}} \sum_{k=1}^{n} m_k.$$

Since V is bounded,

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \le d_{\mathcal{W}}(W_n, \Sigma_\infty^{1/2} Z) + O(n^{-1/2}).$$

In stationarity,

$$E(\Sigma_k) = \Sigma_\infty, \qquad V_k = (n-k+1)\Sigma_\infty,$$

so the Gaussian target in Theorem 2 is exactly $\Sigma_\infty^{1/2} Z$.

The martingale increments are uniformly bounded, and their moments can be bounded uniformly over $\beta \in (0,1)$. Moreover,

$$\tilde{C}_1(d,\beta) \le \frac{C}{1-\beta} \quad \text{and} \quad \sum_{q=1}^{n} q^{-(1+\beta)/2} \le \frac{C}{1-\beta} n^{(1-\beta)/2}.$$

It follows that the first Taylor remainder in Theorem 2 is at most

$$C(1-\beta)^{-2} n^{-\beta/2}.$$

For the second Taylor remainder,

$$D_k = \frac{1}{n-k+1} I, \qquad \operatorname{Tr}(D_k) = \frac{d}{n-k+1},$$

and the same bound follows.

It remains to control Δ_n. Let

$$B(x) := \Sigma_\infty - v(x).$$

Since $\pi(B) = 0$, there is a bounded matrix-valued solution Φ of

$$\Phi - P\Phi = B.$$

We may normalize this solution so that $\pi(\Phi) = 0$. Define

$$\Gamma_k := \Phi(X_k) - E[\Phi(X_k) \mid \mathcal{F}_{k-1}].$$

Then

$$E[\Gamma_k \mid \mathcal{F}_{k-1}] = 0,$$

and the Markov property and the Poisson equation give

$$\begin{aligned} B(X_{k-1}) &= \Phi(X_{k-1}) - P\Phi(X_{k-1}) \\ &= \Phi(X_{k-1}) - \Phi(X_k) + \Gamma_k. \end{aligned}$$

Because the chain is stationary,

$$E(\Sigma_k) = \Sigma_\infty, \qquad \Sigma_k = v(X_{k-1}).$$

Hence

$$E(\Sigma_k) - \Sigma_k = \Phi(X_{k-1}) - \Phi(X_k) + \Gamma_k. \tag{5}$$

For fixed $h \in Lip_1$, let

$$H_k := V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2}.$$

Since H_k is $\mathcal{F}_{k-1}$-measurable,

$$E \operatorname{Tr}(H_k \Gamma_k) = 0.$$

Using (5) and summation by parts,

$$\begin{aligned}
&\sum_{k=1}^{n} E\,\mathrm{Tr}\left(H_k(E(\Sigma_k)-\Sigma_k)\right)\\
&= E\,\mathrm{Tr}(H_1\Phi(X_0)) - E\,\mathrm{Tr}(H_n\Phi(X_n))\\
&\quad + \sum_{k=2}^{n} E\,\mathrm{Tr}\left((H_k-H_{k-1})\Phi(X_{k-1})\right).
\end{aligned}$$

The bound on $A_{k,h}$ in Theorem 2 and the identity $V_k = (n-k+1)\Sigma_\infty$ imply

$$\|H_k\|_{op} \le C(n-k+1)^{-1/2}.$$

Since Φ is bounded, the two boundary terms are $O(1)$, uniformly over $h \in Lip_1$. Lemma 2 gives

$$\sup_{h\in Lip_1} \sum_{k=2}^{n} E\|H_k - H_{k-1}\|_{op} \le \frac{C}{(1-\beta)^2} n^{(1-\beta)/2}.$$

Consequently,

$$\begin{aligned}
\Delta_n &\le \frac{C}{\sqrt{n}}\left(1 + \frac{1}{(1-\beta)^2} n^{(1-\beta)/2}\right)\\
&\le \frac{C}{(1-\beta)^2} n^{-\beta/2}.
\end{aligned}$$

Combining the reward boundary estimate, the two Taylor-remainder estimates, and the bound on Δ_n proves the first assertion. Choosing

$$\beta = 1 - \frac{2}{\log n}$$

for all sufficiently large n proves the second. □

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